

PHYSIOLOGICAL RESPONSES OF RED MAPLE SAPLINGS TO SUB-IRRIGATION WITH AN UNTREATED MUNICIPAL LANDFILL LEACHATE

S. C. Shrive and R. A. McBride¹

Department of Land Resource Science, University of Guelph, Guelph, Ontario N1G 2W1, Canada

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An experiment was undertaken to examine the response of hydroponically-grown red maple (*Acer rubrum* L.) saplings to a series of four flooding (sub-irrigation) treatments distributed over a 25-day period with an untreated (saline) municipal solid waste landfill leachate or deionized water. Net photosynthesis rates measured for water-treated saplings rapidly declined to 62% of the levels measured in untreated (control) saplings, but returned to pre-treatment levels with subsequent flooding treatments. Net photosynthesis rates measured for leachate-treated saplings decreased to about 50% of the levels measured for control saplings over the 25-day treatment period, and remained suppressed. Loss of turgor in leaves and a iron-oxyhydroxide plaque on root surfaces were also observed. Reasons proposed for this acute and apparently irreversible response to leachate exposure include: (i) extreme root anaerobiosis conditions caused by root system flooding and exacerbated by a high chemical oxygen demand leachate; (ii) increased root-soil interface resistance to transpiration water flow (osmotic potential gradient, iron oxyhydroxide plaque); (iii) metabolic intolerance to high solute concentrations in plant tissue; and (iv) exposure to potentially toxic volatile organic compounds. Water sub-irrigation had virtually no effect on nutrient and non-nutrient element concentrations in foliage or on the spectral reflectance characteristics of the leaves. Leachate treatment decreased the foliar content of many plant macro- and micro-nutrients significantly, and shifts in spectral reflectance patterns indicated declining plant vigour. Certain chemical constituents present in high concentrations in the leachate irrigant and which can be phytotoxic, such as Cl, accumulated to a significant degree in leachate-treated plant tissue.

Key Words—Municipal solid waste (MSW) landfill leachate, sub-irrigation, net photosynthesis, stomatal conductance, hydroponic, spectral reflectance, plant macro- and micro-nutrients.

1. Introduction

The chemistry of municipal solid waste (MSW) landfill leachate is very complex and variable, and the toxicity of these wastewaters to plants and animals is poorly understood. Nevertheless, phytotoxicity has been cited as a factor in the decline and mortality of trees irrigated with landfill leachates (Menser *et al.* 1983, Ferm 1985). High technology wastewater treatment systems [e.g. reverse osmosis (RO)/ultra-filtration systems] would be capable of removing the offending contaminants from solution or suspension as a

¹ Author to whom correspondence should be addressed.

pre-treatment step before irrigation once they are identified, but most of these technologies are costly and beyond the economic means of many communities (Shrive *et al.* 1994). Other more proven and less costly primary treatment systems (e.g. bioreactors) would not be effective in reducing the ionic strength of saline leachates.

It is widely held that the levels of certain potentially toxic constituents [e.g. ammonia, volatile organic compounds (VOCs)] and conventional contaminants [e.g. chemical oxygen demand (COD)] can be substantially reduced in the irrigants by adopting application methods such as spray irrigation, but direct contact of the leachate with foliage and the resulting residue accumulations are an undesirable side-effect (Shrive *et al.* 1990). Direct uptake of certain toxic VOCs (e.g. trichloroethylene) by plant leaves may also be substantial (Travis & Hattemer-Frey 1988), and this impact on vegetation could extend beyond the leachate treatment area boundaries. Reverse osmosis/membrane technologies are particularly effective in treating saline leachates, but the RO concentrate must still be disposed of in some manner. If land treatment and disposal of MSW leachate is to be implemented with the intent of sustaining or enhancing the health of the recipient terrestrial ecosystem, the chemical properties of the wastewater must be matched with the most appropriate application methods available, which might include surface trickle irrigation or sub-irrigation (Shrive *et al.* 1994).

Acute stress on vegetative physiological processes, caused by practices such as leachate irrigation, can often be recognized and characterized through comparison with known and more naturally occurring stress responses (Landsberg 1986). This study had the objective of characterizing and contrasting the physiological responses of a common northern temperate hardwood species to flooding (sub-irrigation) of the root system with: (i) an untreated, saline MSW leachate; or (ii) deionized water. Tree saplings were subjected to this extreme treatment after a period of hydroponic maintenance on a nutrient solution. The results should provide information on the nature and severity of stresses that might be expected in vegetation exposed to MSW landfill leachates. The findings should be useful to terrestrial ecologists involved in the design of environmentally sound land treatment/disposal systems for MSW leachates where application is to be carried out by: (i) brief, heavy irrigation events based on rapid rate infiltration or overland flow (Zirschky *et al.* 1986); or (ii) slow rate infiltration by sub-irrigation.

2. Materials and methods

2.1 Experimental material preparation

Bare root, red maple (*Acer rubrum* L.) saplings (1 year old) were obtained in April from a provincial tree nursery and stored at 4°C with their roots in moist peat. Red maple is indigenous to southern Ontario and was chosen for its tolerance to wet soil moisture regimes (Whitlow & Harris 1979). In early May, dormancy was gradually broken over a 2-week period and the saplings were then planted in 28-l plastic pots (four saplings per pot) containing 16-mesh silica sand.

The pots were arranged under an automated irrigation system in an atmospherically-controlled glasshouse. Mist sprayers, positioned 10 cm above the pot surfaces, distributed a nutrient solution (Table 1) as described by Hewitt (1966), to the entire potting medium surface but avoided direct contact with sapling leaves. The pots were irrigated daily to near pore capacity and allowed to drain freely, equilibrating at a volumetric moisture content of about 0.31 m³ m⁻³ within 30 min.

TABLE 1
Chemical composition of the treatment solutions

Chemical constituent*	Guelph MSW leachate	Nutrient solution
Cl	464	3.5
NH ₄	865	0
NO ₃	0	168
PO ₄	1.38	41
SO ₄	49	48
K	1301	156
Ca	883	160
Mg	525	36
Na	1109	33
Fe	1.11	2.8
Mn	5.07	0.55
Zn	0.03	0.065
Cu	0	0.064
B	8.21	0.54
Ni	0.52	0
Cd	0.05	0
TOC†	3774	—
pH	7.01	6.20

* In units of mg l⁻¹, with the exception of pH.

† Total organic carbon not determined for nutrient solution.

Air temperatures and relative humidities in the glasshouse ranged between 32°C and 40% (day)–20°C and 75% (night), respectively. After leaf flush and about 2 months of growth under these hydroponic conditions, the saplings were moved to an area outside the glasshouse and exposed to natural environmental conditions. The saplings were allowed to acclimate for 1 week before the first of the leachate flooding treatments commenced. For a given pot, the nutrient solution irrigation was continued daily until the sub-irrigation treatment schedule commenced, and the saplings were destructively sampled. The ranges of air temperature and relative humidity during the conditioning and flooding treatment periods were comparable to those measured in the glasshouse during the early season growth period, although the level of photosynthetically active radiation (PAR) was higher by as much as 100%.

2.2 Experimental design

Randomly selected pots of saplings were subjected to up to four consecutive flooding (sub-irrigation) treatments with untreated MSW leachate (Tables 1 & 2) or deionized water over a period of 25 days (28 July–22 August). A volume of 400 l of untreated leachate was obtained from the City of Guelph sanitary landfill immediately prior to commencement of the first flooding treatment. The solution chemical analyses listed in Table 1 were performed by several analytical methods as described in Bolton & Evans (1991). The VOC analyses in Table 2 were carried out by GC-mass spectroscopy at the National Water Research Institute in Burlington, Ontario.

The flooding treatments were carried out by placing the pots in a large open tank of deionized water or MSW leachate. A new quantity of deionized water was obtained

TABLE 2
Volatile organic compound concentrations of the leachate from samples taken during the first treatment and after 25 days (Treatment 4)

Volatile organic compound*	Treatment 1 ($\mu\text{g l}^{-1}$)	Treatment 4 ($\mu\text{g l}^{-1}$)
Vinyl chloride	17.93	0
Diethylether	177.20	58.04
1,1-dichlorethane	86.16	0
Benzene	15.41	<10
Toluene	160.90	20.61
Ethylbenzene	26.44	<10
<i>m</i> -, <i>p</i> -xylene	40.93	<10
<i>o</i> -xylene	27.53	<10
Trimethylbenzene	<10	0

* Other volatiles, if present, were at levels below detection limits.

for each of the four main flooding treatments, but the same sample of leachate was used throughout and it remained in a rain-sheltered tank that was open to the atmosphere. The irrigant infiltrated the potting medium from the bottom through the pot drainage holes, effectively saturating the root zone. The pots were removed from the tanks when the irrigant began to pond at the surface of the potting medium (about 1 min), and were then allowed to drain freely.

Each of the four main flooding treatments consisted of immersing five pots in leachate (five-fold replication) and three pots in deionized water (three-fold replication), with one pot (four saplings) representing a single treatment replication. All pots eventually used for leachate treatments 2, 3 and 4 were flooded in the freshly collected leachate at the same time for the first treatment, and so on. The four deionized water treatments were carried out in the same way. This reduced any measurement bias that might result from the treatment order, because the pots were not allocated to a particular treatment prior to the experiment. Saplings maintained for the duration of the experiment on nutrient solution only were used as "controls", with a different group of five pots (20 saplings) measured on each of three dates (Table 4).

Destructive sampling of the saplings was carried out once the sub-irrigation treatment schedule for a given pot was completed. This was done to inspect the root systems visually, and to obtain leaf samples for chemical analysis of plant tissue.

2.3 Foliar gas flux densities and related parameter measurement

A LI-COR 6000 portable photosynthesis system (LI-COR Inc., Lincoln, NE) was used to measure foliar carbon dioxide and water vapour flux densities. The instrument calculates the net photosynthesis rate (P_N), transpiration rate (E), leaf internal CO_2 concentration (c_i) and stomatal conductance (C_s) based upon changes in CO_2 concentration and relative humidity in a 1-l polycarbonate cuvette which is clamped directly onto the leaf, creating a closed system environment. These measurements were taken on three mature, fully expanded leaves located within four nodes of the apical meristem on each sapling, giving a total of 12 readings for each treatment replicate. Thus for each main flooding event, a total of 60 and 36 readings were made for the leachate

TABLE 3

Independent atmospheric parameters for the treatment measuring dates over the study period

Sub-irrigation event			Maximum air temperature (°C)	Maximum VPD (kPa)	c_a^* $\mu\text{l l}^{-1}$
No.	Period	Date			
1	Post-irrigation	28 Jul.	30.0	2.88	$328 \pm 8\text{bcd}$
2	Pre-irrigation	29 Jul.	32.6	3.17	$334 \pm 18\text{abc}$
	Post-irrigation	30 Jul.	27.3	2.70	$328 \pm 12\text{bcd}$
3	Pre-irrigation	7 Aug.	25.5	2.71	$321 \pm 7\text{e}$
	Post-irrigation	8 Aug.	30.5	2.79	$335 \pm 16\text{ab}$
4	Pre-irrigation	21 Aug.	21.1	2.47	$334 \pm 12\text{ab}$
	Post-irrigation	22 Aug.	21.5	2.28	$330 \pm 10\text{bc}$

* Mean ambient CO_2 concentration and standard deviation (means followed by different lower case letters are significantly different at $P < 0.05$).

VPD = Vapour pressure deficit.

TABLE 4

Results from simple regressions of net photosynthetic rate upon stomatal conductance for nutrient solution control (N), water-treated (W) and leachate-treated (L) saplings over the study period

Sub-irrigation event			b_0 Intercept*	b_1 Coefficient†	Standard error of estimate ($\mu\text{mol CO}_2$ $\text{m}^{-2}\text{s}^{-1}$)	r^2
No.	Period	Treatment				
1	Post-irrigation	N	1.09 ± 0.01	$0.383 \pm 0.059\text{c}$	1.20	0.308
		W	1.28 ± 0.03	$0.851 \pm 0.051\text{b}$	1.15	0.891
		L	1.14 ± 0.04	$1.53 \pm 0.127\text{a}$	1.30	0.714
2	Pre-irrigation	N	1.20 ± 0.02	$0.626 \pm 0.091\text{b}$	1.16	0.450
		W	1.20 ± 0.03	$0.563 \pm 0.084\text{b}$	1.12	0.567
		L	1.44 ± 0.04	$1.09 \pm 0.058\text{a}$	1.21	0.862
	Post-irrigation	W	1.23 ± 0.02	$0.688 \pm 0.049\text{b}$	1.12	0.854
		L	1.22 ± 0.03	$0.765 \pm 0.065\text{a}$	1.20	0.704
3	Pre-irrigation	W	1.11 ± 0.02	$0.672 \pm 0.071\text{a}$	1.12	0.726
		L	1.07 ± 0.05	$0.580 \pm 0.116\text{a}$	1.29	0.301
	Post-irrigation	W	1.11 ± 0.02	$0.683 \pm 0.069\text{a}$	1.15	0.741
		L	1.17 ± 0.03	$0.595 \pm 0.061\text{b}$	1.19	0.620
4	Pre-irrigation	N	1.07 ± 0.01	$0.271 \pm 0.080\text{c}$	1.20	0.167
		W	1.00 ± 0.01	$0.299 \pm 0.086\text{b}$	1.15	0.263
		L	1.21 ± 0.02	$0.647 \pm 0.041\text{a}$	1.18	0.810
	Post-irrigation	W	0.92 ± 0.02	$0.269 \pm 0.100\text{b}$	1.17	0.175
		L	1.02 ± 0.03	$0.661 \pm 0.079\text{a}$	1.34	0.548

* All regressions significant at $P < 0.05$ (data log-transformed).

† b_1 coefficients followed by different lower case letters within event number and period are significantly different at $P < 0.05$.

and deionized water irrigant treatments, respectively. Similar measurements were made concurrently on the control saplings.

Preliminary testing with the LI-COR 6000 instrument indicated that sapling response

stabilized within 15 min of the flooding treatment, so this adjustment period was routinely allowed to elapse before the actual readings were taken. Measurements were taken randomly amongst the treatment replicates during cloud-free periods between 0900 and 1430 h EST when PAR ranged from 1500–2300 $\mu\text{mol m}^{-2}\text{s}^{-1}$. Tests carried out before the main flooding experiment indicated that the corresponding photosynthetic photon flux density (PPFD) exceeded photosynthetic light saturation for the red maple saplings. A LI-COR 190 quantum sensor was used for all PAR measurements. Independent measurements of air temperature and relative humidity were made with a sling psychrometer.

2.4 Foliar tissue and treatment solution analyses

Four leaves were excised from the upper five nodes of randomly selected saplings subjected to flooding treatments 1, 3 and 4 with the leachate ($n=20$) and deionized water ($n=12$) irrigants. The samples were collected on the day following the flooding treatment. In addition, 24 leaf samples were collected from randomly selected control saplings.

The leaf tissue was dried at 80°C for 48 h and then ashed at 450°C. The residue was dissolved in 1 M HCl and the extract was analysed for K, Na, Ca, Mg (atomic absorption spectroscopy), total N (Kjeldahl) and P (Technicon autoanalyser colorimetry). In addition, the extracts were analysed for trace element content by inductively coupled plasma-atomic emission spectroscopy (ICP-AES). Non-complexed Cl was determined separately by sonifying the ashed residue in solution and analysing the extract by autoanalyser colorimetry.

Samples of the solution that continued to drain slowly from the potting medium were taken from treatment replicates on the day following the flooding treatment. These solutions were analysed for COD using the method outlined by Knechtel (1978), and for osmotic potential (ψ_o) using a thermocouple psychrometer (P.W.S. Instruments Inc., Guelph, ON) in a constant temperature sample chamber (Decagon Devices Inc., Pullman, WA). Solution ψ_o is strongly correlated to the more common measure of electrical conductivity (Jurinak 1990), but ψ_o is the preferred measure of ionic strength for studies involving vegetation and given the complex chemistry of MSW leachates (Cureton *et al.* 1991). These analyses were also performed on the nutrient solution, the untreated leachate, and the excess pore water draining from the pots containing the control saplings.

2.5 Spectroradiometric measurements

Spectroradiometric measurements of leaf adaxial hemispherical reflectance (400–1100 nm as compared to a barium sulphate standard) were made with a LI-COR 1800 spectroradiometer fitted with an integrating sphere, on the day following the flooding treatments. Measurements were made on 30 mature, fully expanded (third internodal) leaves of saplings, randomly selected from each of flooding treatments 1, 3 and 4 in water and leachate. Similar measurements were made on leaves of the control saplings.

3. Results

3.1 General experimental conditions

Atmospheric variables that were measured over the study period and that influence gas flux densities in plant leaves are given in Table 3. Maximum daytime air temperatures and vapour pressure deficits (VPD) were directly related ($r=0.933$) on the seven measurement dates. Mean ambient CO_2 levels varied between $321\text{--}335\ \mu\text{l l}^{-1}$.

Mean osmotic potentials of the excess pore solutions collected from the pots after each irrigation event were significantly lower ($P<0.001$) for the leachate treatments (range $-690\text{--}-820\ \text{kPa}$) in comparison to the deionized water treatments (range $-110\text{--}-180\ \text{kPa}$), where the nutrient solution dominated the pore water chemistry. Drainage solution COD varied correspondingly with ionic strength, reaching up to $10\ \text{g kg}^{-1}$ in the leachate treatments but remained consistently less than $25\ \text{mg kg}^{-1}$ in the water treatments and nutrient solution controls. There was no significant within-treatment variation in pore solution COD or osmotic potential over the period encompassing the four flooding treatments.

3.2 Physical sapling response

Saplings immersed in water or leachate showed no visible physical response during the first flooding (sub-irrigation) period of about 1 min. Within 5 min of the commencement of pot draining, however, leaves from leachate-irrigated plants were beginning to lose turgor and had noticeably darkened in colour. Turgor loss was greatest in the upper nodal leaves and appeared to stabilize within about 15 min. The transpiration rate (E) dropped to between 67–75% of the mean control sapling E after the first treatment, but declined even further to between 50–67% after the subsequent treatments. These symptoms were not observed in the leaves of the water-irrigated plants.

Twenty-four hours after the first leachate treatment, some turgor had returned to leaves and turgor recovered even more following leachate flooding events 2 and 3, although at no time did it fully return to pre-experimental levels. Severe turgor loss occurred in leachate-treated plant leaves within 24 h after the final flooding event. Over the 25-day treatment period, the leaves of leachate-treated plants developed areas of chlorotic and necrotic tissue, but the extent and severity of these symptoms were highly variable. Everard & Drew (1989) observed both turgor loss and premature leaf senescence in sunflowers (*Helianthus annuus* L.) subjected to root system oxygen deficiency. Leaves of water-treated plants showed no visible changes in turgor or colour.

By flooding event 4, destructive sampling of the potted plants showed that a reddish-brown plaque covered much of the upper portions of the root system in the leachate-treated pots. Precipitate coatings with qualitatively similar characteristics have been observed on roots of water oak (*Quercus nigra* L.) and green ash (*Fraxinus pennsylvanica* Marsh.) after prolonged flooding (Good & Patrick 1987), and on roots of cattails (*Typha latifolia* L.) exposed to Fe^{2+} under reducing conditions in solution culture (Taylor *et al.* 1984). The composition of this plaque was believed to be an iron oxyhydroxide (Chen *et al.* 1980, Macfie & Crowder 1987). Large amounts of ferrihydrite, a type of iron oxyhydroxide, have been found in MSW leachates with high Fe concentrations (Shrive *et al.* 1990, Bolton & Evans 1991). Plaque formation on roots has been attributed to the oxidation of Fe^{2+} in the rhizosphere (Chen *et al.* 1980), and may hinder water and nutrient uptake by affected roots over the longer term. Plaque

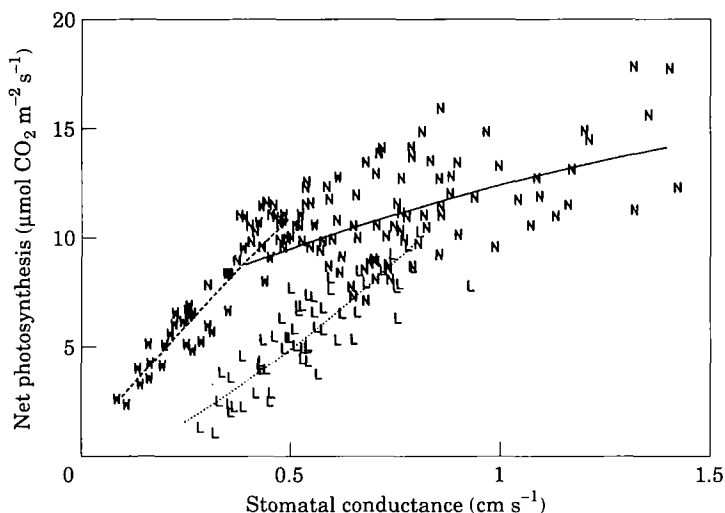


Fig. 1. Net photosynthesis rate response to stomatal conductance after one sub-irrigation treatment with water or leachate. N, nutrient (solid line); W, water (dashed line); L, leachate (dotted line).

was not observed on red maple roots exposed to water sub-irrigation or the maintenance nutrient solution in this study.

Adventitious roots of up to 4 cm in length had developed on leachate-treated plant stems by the fourth and final flooding event. Formation of adventitious roots has been reported on red maple stems after 11 uninterrupted days of root zone flooding (Tripepi & Mitchell 1984). Their development is thought to be a plant response to the need to expand the root system into more aerobic areas of the rooting medium, and may act as an auxiliary pathway for nutrient uptake when normal functioning of the lower root system is hindered by adverse soil conditions (Kozłowski 1982, Nadkarni & Primack 1989). Water-treated sapling stems showed minimal and thinly distributed adventitious root development after four flooding events, and none were observed on the control saplings over the entire experimental period.

3.3 Photosynthesis and stomatal conductance

Mean P_N rates measured for red maple saplings subjected to one sub-irrigation event with deionized water rapidly declined to 62% of the levels measured for control saplings, but gradually returned to pre-treatment levels even with subsequent flooding treatment. The mean P_N rates for leachate-treated saplings decreased to and remained at levels 50% of those for the control saplings throughout the 25-day period. Similar depression of photosynthetic rates has been noted in silver maple (*Acer saccharinum* L.) seedlings where the root systems were submersed in water containing suspended soil particles, as compared to similar treatment with pure water (Peterson & Bazzaz 1984).

Figures 1–4 show the relationships between P_N and C_S for red maple leaves observed on the day before (except Fig. 1) and immediately after flooding events 1–4, respectively, in either water or leachate. Similar functions are plotted for the control sapling leaves in several instances for comparative purposes, and these generally show a lack of response in P_N to increasing C_S . Under these conditions, carboxylation activation of

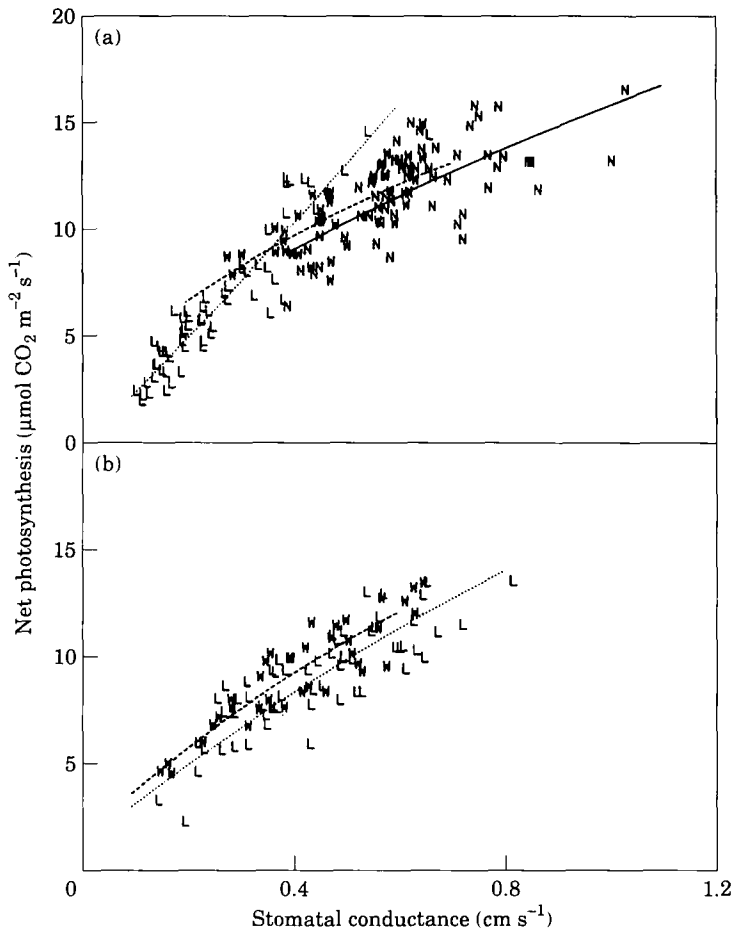


Fig. 2. Net photosynthesis rate response to stomatal conductance after two sub-irrigation treatments with water or leachate. Measurements are shown for the day prior (a) and subsequent (b) to the second flooding treatment event. N, nutrient (solid line); W, water (dashed line); L, leachate (dotted line).

the enzyme ribulose 1, 5-bisphosphate carboxylase-oxygenase (Rubisco) has been maximized (Farquhar & Sharkey 1982). The main limiting factor in photosynthesis at maximum stomatal conductance therefore becomes the capacity to regenerate ribulose bisphosphate.

The data from flooding event 1 (water and leachate), however, showed that P_N and C_S are positively correlated (Fig. 1). Factors other than the natural regenerative capacity of ribulose bisphosphate clearly had a significant negative influence on carbon assimilation in these treatments. The most probable of these factors for the water treatment would be oxygen deficiency for root respiration. Root anaerobiosis may also have been a factor in the photosynthetic response of the leachate-treated saplings (Fig. 1), but the dramatic loss of leaf turgor and other visual symptoms point to deficit moisture stress caused by the depressed osmotic potential of the flooding solution. Foliar uptake of VOCs may also have been involved in the significant downward displacement of the leachate $P_N(C_S)$ relationship in Fig. 1 (Travis & Hattemer-Frey 1988).

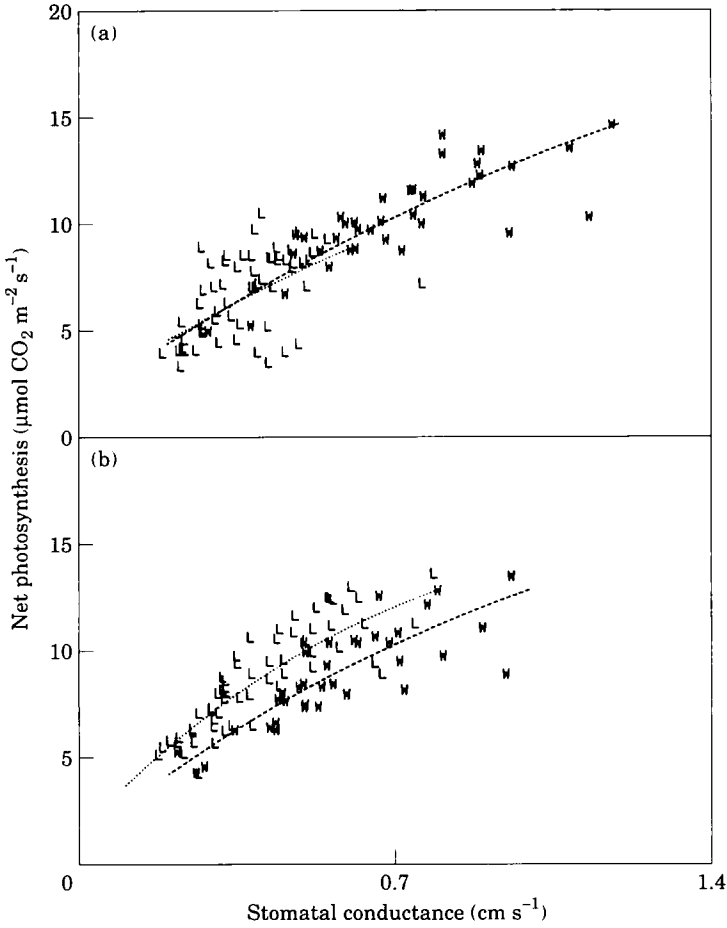


Fig. 3. Net photosynthesis rate response to stomatal conductance after three sub-irrigation treatments with water or leachate. Measurements are shown for the day prior (a) and subsequent (b) to the third flooding treatment event. W, water (dashed line); L, leachate (dotted line).

The P_N rate generally increased in a slightly curvilinear manner with increasing C_s for all treatments (Figs 1–4). Therefore, these data were log-transformed, in order to facilitate between-treatment analysis by least squares analysis (Table 4). The regression coefficient representing the slope (b_1) of the $P_N(C_s)$ relationship decreased significantly ($P < 0.05$) for flooding event 1, in order from leachate-treated to water-treated to the control saplings. In subsequent sub-irrigation events and periods, the differences between the b_1 coefficients for the water and control treatments were either less pronounced or not statistically significant. The b_1 coefficients for the leachate treatment, however, remained significantly higher than those of the control and water treatments, except for flooding event 3. The adverse effect of the factor(s) implicated above on the photosynthetic capacity of leachate-treated saplings clearly persisted through the entire 25-day experimental period. Clough & Sim (1989) found a similar trend in slopes of the $P_N(C_s)$ relationship for mangroves under conditions of increasing soil salinity and atmospheric VPD.

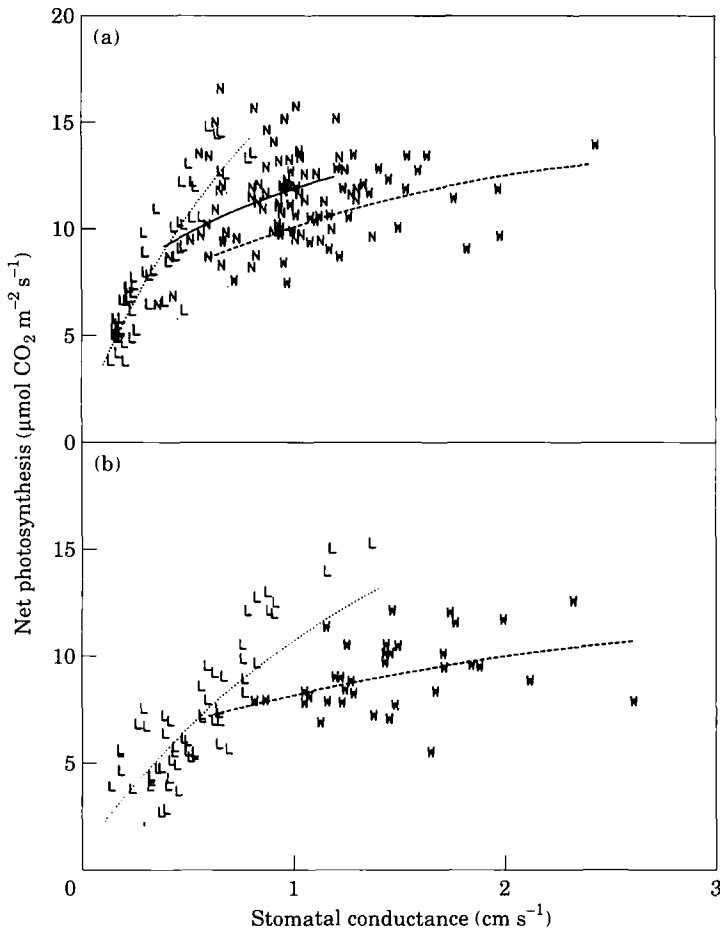


Fig. 4. Net photosynthesis rate response to stomatal conductance after four sub-irrigation treatments with water or leachate. Measurements are shown for the day prior (a) and subsequent (b) to the fourth flooding treatment event. N, nutrient (solid line); W, water (dashed line); L, leachate (dotted line).

Net photosynthesis rates were measured as a function of leaf intercellular CO_2 concentrations (c_i) at ambient CO_2 levels to develop partial $P_N(c_i)$ curves for flooding events 1, 2 and 4 (Fig. 5). These functions are typically developed under a series of external CO_2 partial pressures for use in the measurement of CO_2 compensation points of plants (Bauer & Martha 1981), and for the partitioning of stomatal and non-stomatal limitations to CO_2 uptake (Farquhar & Sharkey 1982, Jones 1985, Assmann 1988). Partial $P_N(c_i)$ curves were derived in this study at ambient CO_2 levels (Table 3) after Lee *et al.* (1989), and were plotted as the mean measured P_N rate calculated for c_i intervals of $20 \mu\text{l l}^{-1}$. The broken lines in Figs 5(a) & 5(b) are normalized $P_N(c_i)$ functions for sugar maple (*Acer saccharum* Marsh.) reported by Hesketh (1963).

The curves in Figs 5(a) & 5(b) were used here to assist in distinguishing treatment-related effects on CO_2 fixation. Like $P_N(C_s)$ functions, the configuration of these relationships is typically asymptotic, thus depicting the regions of Rubisco demand and saturation in relation to c_i (Farquhar & Sharkey 1982). The upper region of the curve

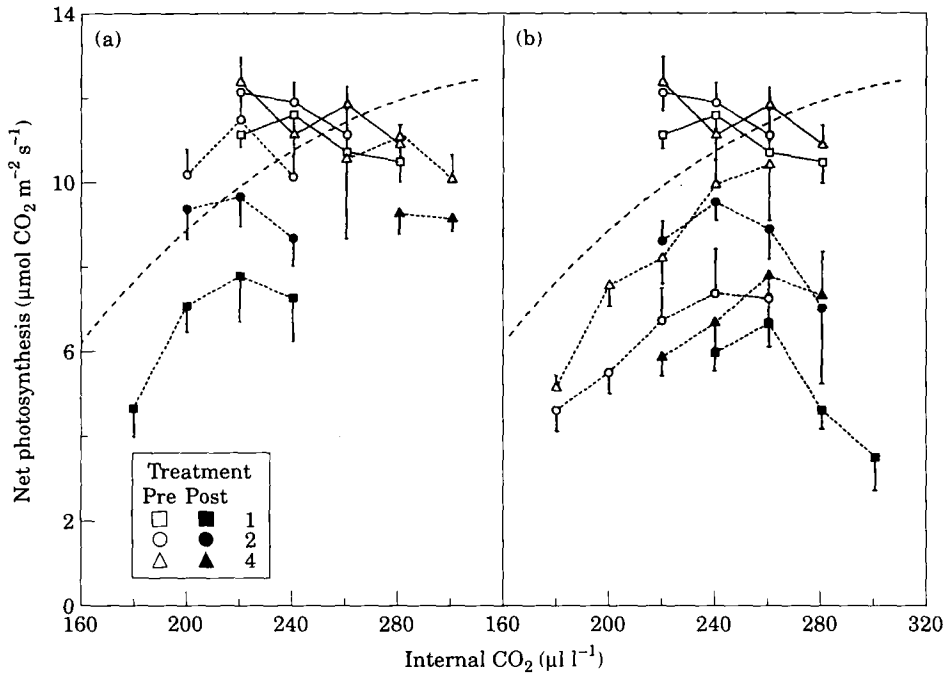


Fig. 5. Net photosynthesis rate response to leaf internal CO_2 after sub-irrigation treatments with (a) water or (b) leachate. Measurements are shown for the day prior and subsequent to treatment events 1, 2 and 4. Error bars are ± 1 standard deviation. Curvilinear function (broken line) is a normalized $P_N(c_i)$ function for sugar maple (after Hesketh, 1963). Nutrient (solid line); Water (dashed line in a); Leachate (dashed line in b).

also represents the “demand net photosynthetic rate” if diffusion resistance to CO_2 uptake is negligible. All assimilation rates measured for the control saplings [Figs 5(a) & (b)] appeared to reach the maximum levels achievable under the experimental conditions and did so within a relatively narrow c_i range. The mean P_N rate for the control plants ranged from $10.9\text{--}11.8\ \mu\text{mol m}^{-2}\text{s}^{-1}$ and did not differ significantly over the 25-day period. Jurik (1986) also observed P_N rates for red maple in the field that were relatively constant during the period between full leaf expansion and senescence.

Following flooding event 1, the P_N rate was dramatically reduced in both water- and leachate-irrigated saplings [Figs 5(a) & (b), respectively] in comparison to the control saplings. This reduction in P_N coincided with a reduction in c_i for the water treatment, but the leachate-treated saplings show unchanged or slightly increased c_i levels. The latter response suggests that increased resistance to CO_2 diffusion was not a major factor in reducing P_N in the leachate-treated saplings. Instead, these saplings appear to have been impaired in their ability to fix CO_2 present in their leaf stomatal cavities, even though there was some enrichment in CO_2 concentration. This gives rise to functions which are more quadratic than asymptotic in configuration. Similar $P_N(c_i)$ responses have been reported by Briggs *et al.* (1986) for red maple under deficit moisture stress conditions.

The water-treated saplings seemed more able to recover during flooding events 2 and 4 [Fig. 5(a)]. The mean P_N rates measured during the pre-treatment periods did not differ significantly ($P < 0.05$) from those observed for the control saplings, but the mean

TABLE 5

Mean foliar element concentrations of nutrient solution control saplings and saplings of sub-irrigation events 1, 3 and 4 of the water and leachate treatments

Chemical element*	Nutrient solution	Treatment					
		Water			Leachate		
		1	3	4	1	3	4
g kg ⁻¹							
N	20.5	20.8	21.6	20.5	20.5	22.1	21.4
P	9.9cd	9.0de	14.7a	15.1a	13.1ab	11.8bc	6.5e
K	7.3	5.3	7.5	7.5	6.3	7.5	5.3
Ca	9.7ab	8.3b	11.2a	9.9ab	10.1ab	9.0ab	5.3c
Mg	3.1	2.8	2.9	2.7	2.6	2.7	2.7
mg kg ⁻¹							
Mn	1807b	883b	2084ab	1460b	5549a	1696b	717b
Fe	58.9a	49.6ab	60.2a	46.3ab	66.9a	46.7ab	30.1b
B	9.94c	6.00c	12.8c	9.06c	20.0b	31.0a	25.8ab
Zn	17.5c	27.5a	25.0ab	14.5cd	19.0bc	14.2c	8.6d
Cd	2.49a	1.56ab	2.27ab	1.57ab	0.93b	2.16ab	2.33ab
Co	0.02c	0.02c	0.24bc	0.38bc	1.09a	0.97a	0.49b
Cu	8.35ab	7.63ab	7.47ab	6.43ab	10.7a	10.4ab	3.53b
Ni	20.9a	20.4a	18.2ab	13.4abc	4.7c	6.1bc	16.3abc
Pb	2.74a	2.35ab	1.54b	1.81ab	1.52b	0.16c	0.11c
Na	62.9	45.6	56.2	47.0	56.6	51.2	47.8
Cl	434c	444bc	506bc	446bc	593bc	606b	959a

* Mean concentrations for a given element followed by different lower case letters are significantly different at $P < 0.05$.

P_N rates were lower in the post-treatment period over a similar c_i range. Less resiliency and increased variability in physiological functioning was evident in the leachate-treated saplings in later irrigation events [Fig. 5(b)]. Mean P_N rates remained suppressed but spanned a wider range of c_i , causing downward displacement of the $P_N(c_i)$ curve. Ball & Farquhar (1984) observed similar declines in $P_N(c_i)$ curves measured for mangrove trees subjected to chronic saline soil conditions. In the present study, red maple (a nonhalophyte) appeared to function at a stress tolerance threshold under the conditions of flooding events 2–4 and was less able to adjust and recover from repeated leachate sub-irrigation.

3.4 Foliar element concentrations

Foliar element concentrations measured for control saplings and for saplings that were subjected to flooding events 1, 2 and 4 (water and leachate) are presented in Table 5. Plant tissue concentrations of N, K, Na, Ca, Mg and the micro-nutrients in the control saplings are consistent with those observed in red maple under field conditions in temperate latitudes, although P was well above sufficient levels (Ricklefs & Matthew 1982, Erdmann *et al.* 1988).

Levels of N, K, Na and Mg did not change significantly ($P < 0.05$) in water- or leachate-treated sapling leaves over the 25-day period when compared to control leaves.

No significant changes were observed in the levels of most other analysed elements in water-treated sapling foliage except for an observed rise in P. There was no leachate treatment effect on foliar N concentration as has been reported elsewhere (Gordon *et al.* 1989a,b), which may be attributed to the maintenance of favourable N availability prior to the flooding treatments (Stanturf *et al.* 1989).

Levels of other elements in leachate-treated sapling leaves were more variable. For example, levels of P and Mn tended to increase and then fall substantially with successive sub-irrigation treatments, whereas levels of Ca and Zn generally declined over the duration of the experiment. Similar patterns of nutrient element concentration response to toxic levels of metals in soils (e.g. Cu, Ni) and to other stress conditions have been reported elsewhere for *Acer* spp. (Lozano & Morrison 1981).

Concentrations of B and Cl more than doubled in leachate-treated sapling leaves compared to the control saplings over 25 days. Both B and Cl can reach toxic levels in plants, but species-specific tissue concentrations are not well established, particularly for woody species (Gupta *et al.* 1985). These results are consistent with other studies examining the effects of MSW leachate irrigation on the foliar concentrations of these elements in *Acer* spp. and other temperate hardwood species (Gordon *et al.* 1989a,b).

3.5 Leaf spectral reflectance characteristics

The spectral reflectance patterns of sapling leaves for the various treatments and sub-irrigation events are shown in Fig. 6. In general, visible band reflectance (400–700 nm) of water-treated sapling leaves decreased over the study period but increased in the leachate treatments. This effect was most pronounced in the green portion of the spectrum (about 550 nm). Other studies have shown that significant inverse relationships exist between reflectance levels at 550 nm and foliar N content (Walburg *et al.* 1982, Hinzman *et al.* 1986). Since foliar N did not vary significantly with treatment or time in this study (Table 5), other treatment-related factors must have been responsible.

Stress-induced changes in the reflectance properties of leaves are also usually first detected in the visible bands, and most notably in the “chlorophyll well” at 660–680 nm, because of chlorophyll content sensitivity to physiological disturbance (Knippling 1970, Boyer *et al.* 1988, Card *et al.* 1988). Increasing reflectances were observed in the chlorophyll well region of leachate-treated sapling leaves over the study period, suggesting a decreased chlorophyll content and therefore a diminished photosynthetic capacity for this treatment. Stresses from heavy metal uptake can also cause increased reflectance at these wavelengths (Horler *et al.* 1980, Schwaller *et al.* 1983), but the data in Table 5 show no clear evidence that this was a contributing factor. Studies by Ashraf (1989) and Seemann & Critchley (1985), however, have shown that a reduction in leaf chlorophyll occurred in salt-stressed nonhalophytes. Reflectances at the 660 and 680 nm wavelengths for the water-treated sapling leaves did not differ significantly from the levels measured for the control saplings over the study period, so this is the most probable stress causing changes in leaf spectral signature.

A parameter, λ_{rc} , has been defined by Horler *et al.* (1983) as the wavelength of maximum slope found in the near infrared area of steep reflectance increase (680–750 nm) known as the “red edge”. The position of λ_{rc} on the spectrum has been shown to be dependent on chlorophyll concentration, with a shifting of both the red edge and λ_{rc} toward greater wavelengths generally coinciding with increasing width of the absorbance well. Remotely sensed imagery of areas undergoing forest decline or vegetation growing

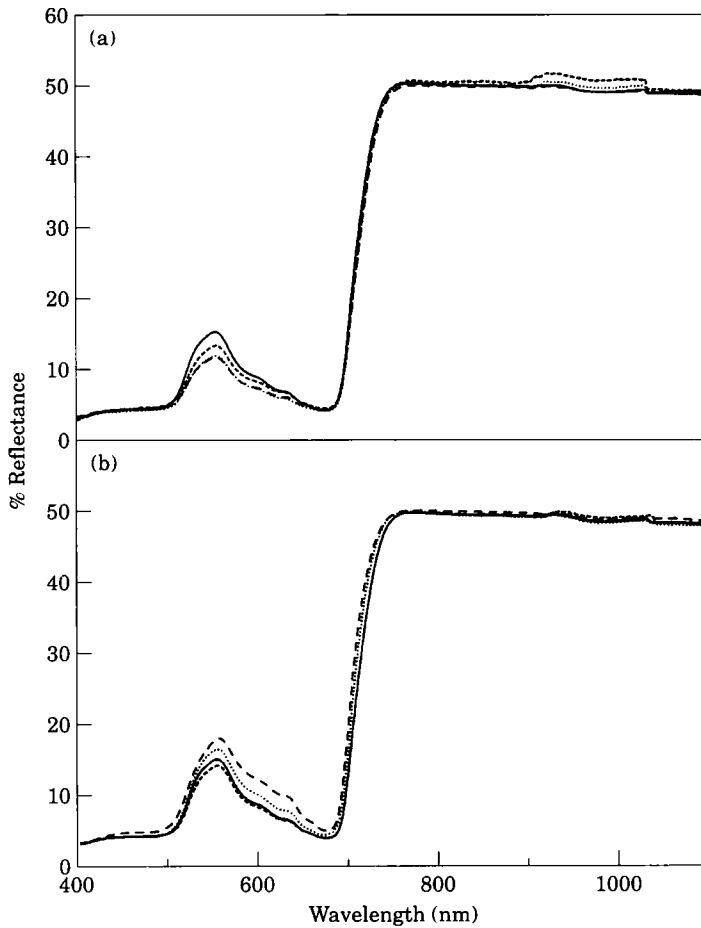


Fig. 6. Leaf spectral reflectance through four sub-irrigation treatments with (a) water or (b) leachate. Nutrient (solid line); Treatment 1 (dashed line); Treatment 3 (dotted line); Treatment 4 (broken line).

in soils overlying metal ore deposits has revealed a shifting towards shorter wavelengths of the red edge and λ_{re} (Collins *et al.* 1983, Singhroy *et al.* 1986, Rock *et al.* 1988).

A significant shift in λ_{re} toward longer (water treatment) and shorter (leachate treatment) wavelengths had occurred by flooding event 4, when compared to the controls (Table 6). The magnitude of the first derivative (slope) at λ_{re} also decreased (water treatment) and increased (leachate treatment) significantly with successive sub-irrigation treatments. These spectral responses are indicative of sustained or enhanced plant health in the water-treated saplings, and of declining plant health in the leachate-treated saplings.

Trends in P_N rate and foliar Cl concentration in relation to λ_{re} (Fig. 7) show a divergent pattern of treatment effects. Foliar Cl levels and P_N rates remained virtually constant in the water-treated saplings while general plant health, represented by λ_{re} , was sustained or slightly enhanced with water sub-irrigation. In contrast, P_N rates in leachate-treated saplings showed an initial large decline, then stabilized while leaves became increasingly chlorotic and showed signs of premature senescence as λ_{re} decreased.

TABLE 6
Red edge parameters and maximum red edge slopes for leaves of nutrient solution control saplings and saplings of sub-irrigation events 1, 3 and 4 of the water and leachate treatments

Treatment	Wavelength of maximum slope*, λ_{re} (nm)	Maximum dR/dWL* (R% nm ⁻¹)
Nutrient solution	703.4bc	1.33cd
Water		
1	706.9ab	1.41bc
3	707.1ab	1.26cd
4	709.7a	1.23d
Leachate		
1	704.9b	1.39c
3	701.5cd	1.53ab
4	699.0d	1.59a

* Parameter means by column followed by different lower case letters are significantly different at $P < 0.05$. R, reflectance; WL, wavelength.

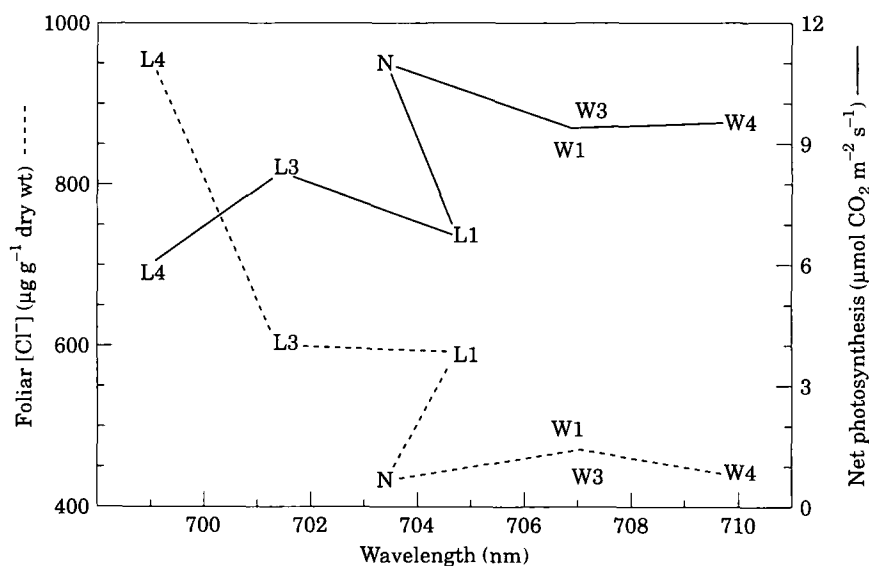


Fig. 7. Foliar Cl⁻ concentration and net photosynthesis rate in relation to λ_{re} for nutrient solution control saplings and saplings subjected to four sub-irrigation events in water or leachate. N, nutrient; W, water; L, leachate; Data label numbers indicate treatment event.

Foliar Cl⁻ concentrations more than doubled in relation to the water-treated and control sapling levels, which is characteristic of nonhalophytes under osmotically-induced deficit moisture stress (Sen-Gupta & Berkowitz 1987).

4. Discussion

It is postulated that the acute physiological stresses observed and measured in the red maple saplings with root system submersion in a saline MSW leachate, were a result of at least three significant changes in the root environment. First, an interfacial resistance to transpiration was likely to have developed as a result of solute accumulation at the root surface or endodermis (Nulsen & Thurtell 1978). Convection of solutes to the root at a higher rate than plant uptake of those solutes can create large osmotic potential differences between the xylem solution and that external to the root (Tinker 1976). This can cause a large source of resistance for the movement of transpiration water, and induce acute deficit moisture stress in the plant as a whole. Plaque formation on the root surfaces may have contributed to this increased resistance at the root-soil interface. Prolonged turgor loss can lead to permanent wilting and plant mortality.

Second, a metabolic intolerance to high ion concentrations in root tissue can result from high ionic strength of soil pore water. If the plant cannot adjust to high salt levels through avoidance mechanisms (i.e. control of uptake) or tolerance mechanisms (i.e. compartmentalization of ions), then ions may concentrate in the cell walls and cytoplasm. The resulting localized cell water deficits will adversely affect metabolism and increase resistance to CO_2 transport in leaf mesophyll cells (Greenway & Munns 1980). In this experiment, the effect of higher ion concentrations (e.g. Cl, Table 5) in plant cells may also have been exacerbated by short-term oxygen deficiency caused by root system flooding treatments. It would be expected that root anaerobiosis would be more extreme in the leachate treatments than in the water treatments, due to the high COD of the Guelph leachate. Together, these effects can impair root activity and lead to conditions of reduced nutrient uptake, phytohormone imbalance and premature leaf senescence over the longer term (Drew 1983). Premature leaf senescence has been observed in tree saplings grown under less harsh conditions of root exposure to MSW leachate (Cureton *et al.* 1991).

Thirdly, it may be important to differentiate between the sapling response to flooding event 1 and the responses to subsequent treatments, because of the marked decreases observed in VOC concentration in the leachate treatment solution over the duration of the experiment (Table 2). It was noted earlier in Fig. 5(b) that the range of c_i over which photosynthesis occurred after flooding event 1 with leachate, was equivalent to or higher than for the water-irrigated (flooding event 1) or control saplings. It is unlikely that the higher c_i levels observed after the first leachate treatment were caused by increased plant respiration given their condition of acute stress. The observed reduction in the P_N rate for the leachate treatment that accompanied the elevated c_i levels may have been caused in part by a decreased demand for CO_2 by Rubisco, brought about by VOC exposure. Smith *et al.* (1976) found that the CO_2 compensation point fell when plants were exposed to solutions containing organic solvents.

There are two main exposure pathways for VOC entry into plant tissue: root uptake from VOCs in solution and foliar uptake from the air (Bacci & Gaggi 1985, Travis & Hattemer-Frey 1988). Once in the root, VOCs can adversely affect transpiration, metabolism and the efficacy of phytohormones but these effects are not well understood. Concentrations of VOCs in tree foliage have been correlated with atmospheric levels (Frank & Frank 1989) and volatile hydrocarbons have been shown to decrease stomatal closure responsiveness (Squier *et al.* 1985). Many of the VOCs found in the treatment leachate rapidly volatilize when mixed with water (Table 2) and exposed to the free atmosphere (Ince & Inel 1989).

5. Conclusions

An experiment was undertaken under a combination of ambient (mid-summer) and controlled-atmosphere conditions to examine the response of hydroponically-grown red maple saplings to a series of four flooding (sub-irrigation) treatments, distributed over a 25-day period with: (i) a saline leachate (untreated); or (ii) deionized water. The principal findings were as follows:

- (1) Net photosynthesis rates measured for water-treated saplings rapidly declined immediately after the first flooding event to 62% of the levels measured in untreated (control) saplings, but returned to pre-treatment levels with subsequent flooding treatments. In this case, oxygen deficiency in the root system (anaerobiosis) probably elicited a hormonal response controlling stomatal conductance and CO₂ fixation.
- (2) Net photosynthesis rates measured for leachate-treated saplings decreased to and remained at about 50% of the levels measured for control saplings over the 25-day treatment period.
- (3) Only the leachate-treated saplings showed visible signs of turgor loss and leaf discoloration (chlorosis, necrosis), symptomatic of osmotically-induced deficit moisture stress from the salinity of the treatment solution and the depressed osmotic potentials (-690 – -820 kPa). This diagnosis was verified by an absence of a measurable transpiration rate for some time after flooding the root system with leachate.
- (4) Sub-irrigation of root systems with deionized water had virtually no effect on the concentrations of nutrient and non-nutrient elements in foliage, or on the spectral reflectance characteristics of the leaves.
- (5) Leachate treatment decreased the foliar content of many macro- and micro-nutrients significantly and induced shifts in spectral reflectance patterns indicating a general decline in plant vigour. The levels of some potentially phytotoxic constituents present in high concentrations in the leachate irrigant (e.g. Cl) rose significantly in plant leaf tissue.
- (6) Elevated leaf internal CO₂ concentrations in leachate-treated saplings suggest that factors apart from stomatal control of CO₂ fixation contributed to the reduced photosynthetic capacity, possibly including metabolic intolerance to high solute concentrations in plant tissue and exposure through leaf or root pathways to potentially toxic VOCs in the leachate.

This experiment has illustrated that acute physiological stresses can be anticipated in young woody vegetation when full-scale land treatment/disposal systems expose that vegetation to saline (untreated) MSW landfill leachates under the type of soil, climate and wastewater application conditions represented in this experimental design. The mitigation of these impacts on vegetation should include irrigation system design (e.g. rapid rate infiltration, overland flow or slow rate infiltration by sub-irrigation) which avoids even short-term, field-saturated soil conditions in the plant root zone.

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